Example 2.24

Find the principal argument $\operatorname{Arg} z$, when $z = \dfrac{-2}{1+i\sqrt{3}}$.

Solution

$$\begin{aligned}
\arg z &= \arg\frac{-2}{1+i\sqrt{3}} \\
&= \arg(-2) - \arg\left(1+i\sqrt{3}\right) \quad \left(\because \arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2\right) \\
&= \left(\pi - \tan^{-1}\left(\frac{0}{2}\right)\right) - \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) \\
&= \pi - \frac{\pi}{3} = \frac{2\pi}{3}
\end{aligned}$$

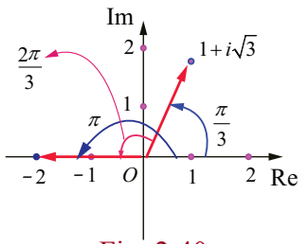


Fig. 2.40

This implies that one of the values of $\arg z$ is $\dfrac{2\pi}{3}$.

Since $\dfrac{2\pi}{3}$ lies between $-\pi$ and π, the principal argument $\operatorname{Arg} z$ is $\dfrac{2\pi}{3}$.

Properties of polar form

Property 1 If $z = r(\cos\theta + i\sin\theta)$, then $z^{-1} = \dfrac{1}{r}(\cos\theta - i\sin\theta)$.

Proof

$$\begin{aligned}
z^{-1} &= \frac{1}{z} = \frac{1}{r(\cos\theta + i\sin\theta)} \\
&= \frac{(\cos\theta - i\sin\theta)}{r(\cos\theta + i\sin\theta)(\cos\theta - i\sin\theta)} \\
&= \frac{(\cos\theta - i\sin\theta)}{r(\cos^2\theta + \sin^2\theta)} \\
z^{-1} &= \frac{1}{r}(\cos\theta - i\sin\theta).
\end{aligned}$$

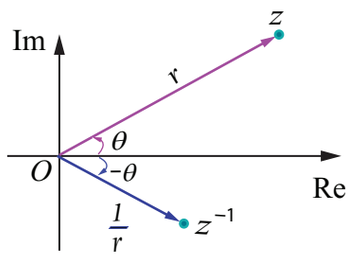


Fig. 2.41

Property 2

If $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$,

then $z_1 z_2 = r_1 r_2(\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2))$.

Proof

$$\begin{aligned}
z_1 &= r_1(\cos\theta_1 + i\sin\theta_1) \text{ and} \\
z_2 &= r_2(\cos\theta_2 + i\sin\theta_2)
\end{aligned}$$

$$\Rightarrow z_1 z_2 = r_1(\cos\theta_1 + i\sin\theta_1) r_2(\cos\theta_2 + i\sin\theta_2)$$

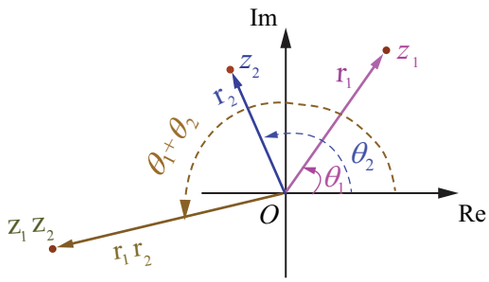


Fig. 2.42

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